

Cadi Ayyad University

Higher School of Technology of Safi

Computer Science Department • Web Techniques

Prof. Soufiane Hourri

Session 6 — Introduction to JavaScript and Client-Side Interaction

Objective: Students understand JS syntax, functions, events, and basic DOM manipulation.

1. Key Concepts

What is JavaScript?

JavaScript (JS) runs **in the browser** after the page loads. It makes pages interactive by responding to user actions (clicks, typing, mouse movement) without reloading the page. Unlike PHP (server-side), JS runs entirely on the client — the user's computer executes it.

Variables, Types, and Operators

```
let name   = 'Alice';           // string
let age    = 20;                // number
let passed = true;              // boolean
const PI   = 3.14;              // constant (cannot be reassigned)

let msg = 'Hello ' + name;      // string concatenation
let area = PI * 5 * 5;          // arithmetic
let check = (age >= 18);        // comparison -> boolean
```

Use `let` for variables that may change. Use `const` for constants. Avoid `var` (old style).

Output Methods

Concept	Meaning
<code>alert('msg')</code>	Pop-up dialog box — blocks the page.
<code>console.log('msg')</code>	Outputs to the browser's developer console (F12).
<code>prompt('question')</code>	Pop-up input box — returns the user's typed value.
<code>document.write('html')</code> <code>)</code>	Writes HTML directly into the document (rarely used).

Functions

```
// Declaration
function greet(name, time) {
    return 'Good ' + time + ', ' + name + '!';
}

// Call
let msg = greet('Soufiane', 'morning');
console.log(msg);    // Good morning, Soufiane!
```

Events and the DOM

The **DOM** (Document Object Model) is the browser's internal representation of the HTML page. JavaScript can read and change DOM elements.

```
<!-- HTML -->
<h2 id="status">Loading...</h2>
<button onclick="updateStatus()">Check</button>

<script>
    function updateStatus() {
        let el = document.getElementById('status');
        el.innerText = 'System is running!';
        el.style.color = 'green';
    }
</script>
```

2. Guided Examples

Example A — Input, condition, and output

```
<script>
    let score = parseInt(prompt('Enter your score (0-20):'));
    let grade;
    if (score >= 16)      { grade = 'A'; }
    else if (score >= 12) { grade = 'B'; }
    else if (score >= 10) { grade = 'C'; }
    else                  { grade = 'F'; }
    alert('Your grade is: ' + grade);
</script>
```

Example B — Function and DOM update

```

<p id="counter">Clicks: 0</p>
<button onclick="increment()">Click me</button>

<script>
  let count = 0;
  function increment() {
    count++;
    document.getElementById('counter').innerText = 'Clicks: ' + count;
  }
</script>

```

3. Exercises

Exercise 1 — Trace the Function Calls

Study the functions and the calls at the bottom. Write the exact output each `console.log` call produces.

```

function power(base, exp) {
  let result = 1;
  for (let i = 0; i < exp; i++) {
    result = result * base;
  }
  return result;
}

function describe(n) {
  if (n % 2 === 0) { return n + ' is even'; }
  else           { return n + ' is odd'; }
}

console.log(power(2, 4));           // Line A
console.log(power(3, 0));           // Line B
console.log(describe(7));           // Line C
console.log(describe(power(2,3))); // Line D

```

Line A output: _____

Line B output: _____

Line C output: _____

Line D output: _____ (explain step-by-step)

Exercise 2 — Write a Complete JavaScript Feature

On paper, write the HTML and JavaScript for a small interactive feature:

A simple temperature converter.

- There is a paragraph with id `result` showing 'Result: --'.
- There is a button labelled 'Convert 100°C to Fahrenheit'.
- When clicked, it calls a function `convert()`.
- The function calculates $F = (C \times 9/5) + 32$ for $C = 100$.
- It updates the paragraph text to show: 'Result: 212°F'.

Write your complete HTML + JavaScript below:

Exercise 3 — Match: Command to Role and Behaviour

For each JavaScript expression in column A, write the letter of the matching description from column B. Then write the expected output or effect.

Column A (expression)	Match	Column B (description)
<code>typeof 42</code>	[]	a) Stores the user's keyboard input in a variable
<code>let x = prompt('?')</code>	[]	b) Returns the string 'number'
<code>parseInt('15px')</code>	[]	c) Selects the element with id 'box'
<code>document.getElementById('box')</code>	[]	d) Converts a string starting with digits to an integer
<code>arr.length</code>	[]	e) Returns the number of items in an array

Exercise 4 — Analyse and Extend

Read the function below and answer all questions.

```
function classify(score) {  
  if (score < 0 || score > 20) {  
    return 'Invalid score';  
  }  
  let grade = Math.floor(score / 4);  
  let labels = ['F', 'F', 'C', 'B', 'A', 'A'];  
  return labels[grade];  
}
```

a) What does `Math.floor(score / 4)` compute for `score = 17`?

b) What does `classify(9)` return? Show each step.

c) What does `classify(21)` return? Why?

d) Extend the function: add a check that converts `score` to a number with `Number(score)` before the if-statement, and returns 'Invalid score' if the result is `NaN`.

Exercise 5 — JavaScript Debugging — Find 5 Errors

The script below contains **5 errors** (syntax, naming, logic, type). For each error, state the line number, what is wrong, and write the corrected line.

```
<script>
  let total = 0;
  const prices = [10, 25, 8, 40];

  for (let i = 0; i <= prices.length; i++) {
    total += prices[i];
  }

  function showTotal {
    document.getElementById('output').innerText = 'Total: ' + total;
  }

  showTotal();
</script>
```

Error 1 (line __): _____

Error 2 (line __): _____

Error 3 (line __): _____

Error 4 (line __): _____

Error 5 (line __): _____

4. Session Recap

JavaScript runs in the browser (client-side) and makes pages interactive. let and const declare variables; functions group reusable logic. if/else handles decisions; for/while repeat actions. alert, console.log, and prompt communicate with the user. The DOM (document.getElementById) connects JavaScript to HTML elements, allowing scripts to read and change content, styles, and structure in real time. JavaScript is the third pillar of web development, alongside HTML and CSS.